MUSHFIQUR RAHMAN SAJID

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Backend-focused Computer Science graduate skilled in **C# | .NET | Python | FastAPI** and relational database design. Competitive programmer with **1,500+** solved problems and experience building **REST APIs** and ML-based systems.

EDUCATION

Ahsanullah University of Science and Technology *Dhaka, Bangladesh*

B.Sc. in Computer Science and Engineering; CGPA: 3.288/4.00 *July 2022 – July 2026*

COMPETITIVE PROGRAMMING & ACHIEVEMENTS / ACTIVITIES

- Solved **1,500+ problems** across : Codeforces | CodeChef | Vjudge | Toph | LeetCode | LightOJ | CSES | AtCoder
- **ICPC Regional Onsite Finalist** BUBT, Dhaka (2025) and DIU, Dhaka (2024).
- Intra AUST Programming Contest - **Champion**, CSE Carnival 4.0 (2024). **2nd Runner-Up**, CSE Carnival 3.0. **3rd Runner-Up**, CSE Carnival 5.0.
- Inter-University Programming Contest (IUPC) – **Top-50** finishes across **7** university contests in 2025 (MU, CUET, DUET, UU, KUET, AUST, UIU) — best result **36th / 100 at MU, Sylhet**.
- Inter-University Collaborative Programming Contest (IUCPC) - **3rd Runner-Up**, BUBT, Dhaka (2025). **Top-20** finishes at UAP (2024) and DRMC Tech Carnival (2024).
- **Problem Setter & Lead Organizer**, Intra AUST Programming Contest (IAPC) 6.0.
- **1st Runner-Up**, Intra AUST Math Olympiad; Participant, 16th National Undergraduate Math Olympiad (NUMO 2025).

TECHNICAL SKILLS

Languages: C, C++, Java, Python, C#

Backend & APIs: ASP.NET Core Web API, FastAPI, REST APIs, SQLAlchemy, Entity Framework Core

Frontend: Angular, JavaScript, HTML, CSS

Databases: Microsoft SQL Server, MySQL, SQL, Relational Database Design, Database Normalization

Authentication & API: JWT, Pydantic, Swagger/OpenAPI

Tools: Git, GitHub, MySQL Workbench, Postman

Cloud & Deployment: Microsoft Azure, Docker, Vercel, Render, Environment-Based Secret Management

Familiar: PHP, OAuth2

Core CS: OOP, Data Structures & Algorithms, Operating Systems, Computer Networking, Database Systems

PERSONAL PROJECT

Mukhosh - *Github* ***https://mukhosh-beta.vercel.app/***

Angular 18, TypeScript, C#, ASP.NET Core (.NET 8), Entity Framework Core, Microsoft SQL Server, Azure SQL Database, Docker, Render, Vercel, k6, Postman

- Designed a **layered REST API** using Controllers, DTOs, Mappers, and Repositories with **EF Core and SQL Server**, modeling relationships across Users, Posts, Comments, Universities, and Reviews.
- Implemented **ASP.NET Core Identity, JWT authentication, role-based authorization (Admin/User), email verification, password reset, and per-IP rate limiting** for authentication endpoints, tested the complete API in **Postman**.
- Load- and concurrency-tested the API with **k6 across 8 scenarios**, achieving **1,750+ automated checks with 0% HTTP errors and up to 50 concurrent users**; identified and fixed a race condition causing duplicate writes.
- **Containerized and deployed** the API with Docker on **Render**, using **Azure SQL Database** for production data and environment variables for secrets and configuration; integrated API-based transactional email for production verification and password resets.

Onnorokom - Backend *Ongoing*

Python, FastAPI, SQLAlchemy, PyJWT, bcrypt, MySQL

- Building a **FastAPI** backend for a university admission/info platform, structuring the codebase into routers, services, repositories, and **Pydantic** schemas to separate request validation, business logic, and database access via **SQLAlchemy.** Implemented **Identity-based authentication** with **JWT** and **role-based authorization** (Admin/User), and enforced business rules such as one-review-per-user via unique composite database constraints.
- Implemented authentication from scratch: **bcrypt** password hashing, custom **JWT** issuing and verification with expiry checks.
- Added centralized exception handling for invalid credentials and expired/invalid tokens mapped to proper HTTP status codes, plus endpoints for profile updates and password changes with old/new password validation.

UNDERGRADUATE THESIS

Continuous Sign Language Recognition in Bangla Using Transformer *June 2026*

1st Runner-Up, AUST CSE Project & Thesis Fest

- Developed a CSLR for Bangla Sign Language into text **MS-TCN and Transformer** architecture, and **BanglaT5**

ACADEMIC PROJECTS

- **Colors** - T-shirt customization tool with real-time preview, built in **Java**.
- **PrintCraft** - web-based custom T-shirt design and order management system, built in **PHP**.